

The response of marine pelagic calcifiers to ocean alkalinity enhancement as a carbon dioxide removal technique

Abstract: It is apparent that in order to reach international climate goals, it will not be sufficient to simply reduce greenhouse gas emissions. There must also be a degree of removal of legacy emissions from our atmosphere to remain within 1.5-2 °C of warming relative to pre-industrial values. Several carbon removal schemes have been proposed which use the ocean as the site of removal, termed marine Carbon Dioxide Removal (mCDR). Meta-analysis of mCDR literature conducted as part of this thesis has revealed that there are six key “elimination” criteria that must be satisfied in order for any mCDR scheme to be viable. There are three main “elimination” criteria which are currently lacking from peer-reviewed mCDR publications: (i) completion of field trials, pilot studies and demonstration projects, (ii) assessment of governance/finance/social aspects of mCDR, and (iii) accurate quantification of CO₂ drawdown/CDR potential. The mCDR scheme of Ocean Alkalinity Enhancement (OAE), which involves raising surface ocean Total Alkalinity (TA) to shift the carbonate buffer system such that it absorbs more atmospheric CO₂, has great potential as an mCDR technique due to its co-benefits and relative advancement compared to other mCDR methods.

However, the growth and calcification response of major pelagic calcifiers to OAE (the coccolithophores and the foraminifera) are as yet unquantified. This thesis has constrained these responses for two key “endmember” species of coccolithophore (*E. huxleyi* and *C. braarudii*) and five key species of planktonic foraminifera (*G. ruber*, *G. siphonifera*, *G. bulloides*, *G. sacculifer* and *O. universa*) using in-vitro experiments. Growth and calcification response to OAE has been found to be species- and taxa-specific, similar to responses to ocean acidification studies. There is evidence that *G. siphonifera*, *G. ruber* and *O. universa* may have an optimum reproductive success at TA ~ 3000 μmol kg⁻¹, but calcification responses of planktonic foraminifera in enhanced TA did not change, although these were subject to high variability. *E. huxleyi* did not change its calcification per cell under simulated air-equilibrated OAE (CO₂-replete conditions), whereas *C. braarudii* decreased its calcification per cell. In simulated non-equilibrated OAE (CO₂-limited conditions), *E. huxleyi* increased its calcification per cell. Even if therefore the risk of calcification per cell increasing for major calcifiers is minimal, thereby providing no re-release of CO₂ to the atmosphere through a calcification feedback, the growth of *E. huxleyi* was fertilised by increased availability of HCO₃⁻ during both air-equilibrated and non-equilibrated OAE scenarios. Therefore, larger populations of this major bloom-forming coccolithophore could contribute to an overall increased amount of calcification and re-release of atmospheric CO₂. The lack of fertilisation of *C. braarudii* by increased supply of HCO₃⁻ reveals a fundamental difference between the “opportunistic” carbon acquisition of *E. huxleyi* and the “static” carbon acquisition mechanism of *C. braarudii*. This has implications for understanding the physiology of the genera they come from, Reticulofenestra (*E. huxleyi*) and Coccolithus (*C. braarudii*). The elemental stoichiometry of *E. huxleyi* and *C.*

braarudii was also constrained during experiments, and it was found that C:N of *E. huxleyi* may increase in non-equilibrated OAE scenarios. This would provide a positive feedback on carbon drawdown during OAE, and merits further investigation with the average C:N:P of pelagic organic matter during OAE.

Further work is needed with mesocosm and mesoscale field trials to constrain the overall growth, calcification, and elemental stoichiometry shifts of the pelagic ecosystem in response to OAE, in a timely manner, so that opportunities to use mCDR responsibly and equitably to reach international climate goals do not pass us by.

[Collapse abstract](#)

Authors



Gill, SJ

Division: MPLS
Department: Earth Sciences
Role: Author

Contributors



Rickaby, R

Institution: University of Oxford
Division: MPLS
Department: Earth Sciences
Role: Supervisor
ORCID: 0000-0002-6095-8419



Henderson, G

Institution: University of Oxford
Division: MPLS
Department: Earth Sciences
Role: Supervisor



James, R

Role: Examiner



Bouman, H

Role: Examiner
ORCID:

0000-0002-7407-9431

Funding Natural Environment Research CouncilFunder identifier: <http://dx.doi.org/10.13039/501100000270>

Bibliographic DetailsDOI: [10.5287/ora-2r9nevgjz](https://dx.doi.org/10.5287/ora-2r9nevgjz)[\(https://dx.doi.org/10.5287/ora-2r9nevgjz\)](https://dx.doi.org/10.5287/ora-2r9nevgjz)

Type of award: DPhil

Level of award: Doctoral

Item Description

Language: English

Deposit date: 2023-02-23

ARK identifier: [ark:/29072/ora_8d2df498127c4341973fc567cf9b1ee3](https://n2t.org/ark:/29072/ora_8d2df498127c4341973fc567cf9b1ee3)

Terms of use

Copyright holder: Sophie Jennifer Gill

Copyright date: 2022

Licence: [Terms and Conditions of Use for Oxford University
Research Archive](#)

Metrics**Views and
Downloads**[About views and](#)

